

1247. A lateral herniation into the intervertebral foramen of the intervertebral disc between the second and third lumbar vertebrae may produce all of the following symptoms except: A
- A. weakened abduction of the hip
 - B. weakened adduction of the hip
 - C. weakened flexion of the hip
 - D. weakened extension of the knee
 - E. diminished sensation to the anterior thigh
- Structures deep to the flexor retinaculum at the ankle include all of the following except: C
- A. posterior tibial artery
 - B. tibial nerve
 - C. tibialis anterior tendon
 - D. flexor hallucis longus tendon
 - E. flexor digitorum longus tendon
- Muscles innervated by the lateral plantar nerve include all of the following except: E
- A. quadratus plantae
 - B. abductor digiti minimi
 - C. all dorsal interossei
 - D. all plantar interossei
 - E. all lumbricales
- An individual with a weak right gluteus medius muscle will: B
- A. compensate by leaning their upper body to their left when standing only on the right lower limb
 - B. demonstrate a lowering of their left side of the pelvis when standing only on the right lower limb
 - C. circumduct (abnormally swing) their right hip when the right lower limb is in the swing phase of walking
 - D. both A and B
 - E. all of the above (A-C)
- 1251 Which of the following cannot produce flexion of the hip: A
- A. gluteus maximus
 - B. sartorius
 - C. rectus femoris
 - D. iliopsoas
 - E. pectineus
- Which muscle(s) has its origin from the calcaneus: A
- A. quadratus plantae
 - B. flexor hallucis brevis
 - C. both A and B
 - D. neither A nor B
1253. Tibialis anterior muscle is active: D
- A. to prevent backward sway when standing
 - B. to produce dorsiflexion of the ankle during the swing phase of walking
 - C. to produce plantarflexion of the ankle when walking
 - D. both A and B
 - E. all of the above (A-C)



Medial rotators of the arm include:

D

- A. infraspinatus
- B. teres minor
- C. both A and B
- D. neither A nor B

Which nerves passing through the arm do not supply any of the brachial muscles:

A

- A. median
- B. musculocutaneous
- C. both A and B
- D. neither A nor B

1225. Structures traveling in the spiral (radial) groove of the humerus include:

A

- A. radial nerve
- B. musculocutaneous nerve
- C. both A and B
- D. neither A nor B

1226. Which muscles supinate the forearm:

C

- A. biceps brachii
- B. supinator
- C. both A and B
- D. neither A nor B

One of your patients is involved in a motorcycle accident in which she is thrown from the cycle, across the handlebars. She strikes the roadway on her right side in such a manner that the head is forced to the left and her right shoulder is forcibly depressed. As a result severe traction is placed on the upper two ventral rami which contribute to the brachial plexus. Which muscle(s) might show paralysis or weakness:

B

- A. scalenus anterior
- B. serratus anterior
- C. both A and B
- D. neither A nor B

1228. Structures seen passing through the quadrangular space of the upper limb include:

C

- A. axillary nerve
- B. posterior humeral circumflex artery
- C. both A and B
- D. neither A nor B

Select the correct statement(s) about the cubital fossa:

A

- A. the supinator and brachialis muscles form the floor of the cubital fossa
- B. the ulnar nerve enters the cubital fossa under the cover of the brachioradialis muscle
- C. both A and B
- D. neither A nor B

The nerve originating from the upper trunk of the brachial plexus include:

C

- A. nerve to subclavius
- B. suprascapular
- C. both A and B
- D. neither A nor B

1210. Select the correct statement(s): A
- A. the intercostobrachialis nerve usually unites with the medial brachial cutaneous nerve
 - B. the cephalic vein lies in the lateral intermuscular septum of the arm
 - C. the long head of the triceps brachii muscle originates from the supraglenoid tubercle of the scapula
 - D. the musculocutaneous nerve terminates as the posterior antebrachial cutaneous nerve
 - E. all of the above (A-D)
1211. Select the incorrect statement: C
- A. the axillary sheath fuses directly to the medial intermuscular septum of the arm
 - B. the median nerve usually lies lateral to the brachial artery in the proximal portion of the arm and medial to the same artery in the distal portion of the arm
 - C. the triceps brachii muscle does not cross the shoulder joint
 - D. the musculocutaneous nerve pierces the coracobrachialis and then lies between the brachialis and the biceps brachii muscles
1212. A 30-year old construction worker had a fifty-pound sack of concrete mix drop two stories and land on his right shoulder while working in the new physical rehabilitation floor at the hospital. He suffered damage to the upper two roots of the brachial plexus. Select the incorrect statement about his condition: D
- A. he would be unable to extend his hand at the wrist
 - B. he would be unable to abduct his arm at the shoulder due to loss of innervation to the deltoid and supraspinatus muscles
 - C. his forearm and hand would be pronated due to loss of innervation to the supinator and biceps brachii muscles
 - D. he would have no cutaneous sensation in his little finger
- Which symptom would not be expected in a patient with trauma to the axilla involving the medial cord of the brachial plexus: A
- A. weakened abduction of the wrist
 - B. loss of sensation along the medial side of forearm
 - C. weakened flexion of the little finger
 - D. loss of sensation along the medial side of arm
- Which of the following nerves would not be expected to contain fibers from the fifth cervical nerve: A
- A. medial pectoral nerve
 - B. suprascapular
 - C. long thoracic
 - D. dorsal scapular
- An atherosclerotic plaque has slowly built up a stenosis (a narrow region of the vessel lumen restricting blood flow) of the brachial artery at its bifurcation into the ulnar and radial arteries. Select the most reasonable route(s) of collateral circulation around the stenosis: B
- A. the posterior interosseous recurrent artery to the posterior interosseous artery
 - B. the radial collateral artery to the radial recurrent artery
 - C. the superior ulnar collateral artery to the anterior interosseous artery
 - D. the superior ulnar collateral artery to the middle collateral artery
 - E. all of the above (A-D)

1200. Select the correct statement(s) about the hand: E
- A. the deep palmar arch lies deep to the tendons of the flexor digitorum profundus
 - B. the lumbricals are located in the midpalmar compartment
 - C. the median nerve and its branches are located in the midpalmar compartment
 - D. the midpalmar cleft is a potential space between the palmar interossei and the tendons of the flexor digitorum profundus on the medial side of the palm
 - E. all of the above (A-D)
1201. The ulnar nerve was damaged by a deep laceration at the left wrist when a 17-year old male put his hand through a plate glass window. Select the correct statement(s) about the patient's symptoms: E
- A. he can not adduct his ring finger and hold a pieces of paper between his ring and middle fingers
 - B. the motor innervation to the flexor digiti minimi brevis and the 3rd and 4th lumbricals is lost so that the ring and little finger are hyperextended at the metacarpophalangeal (M-P) joints
 - C. he is able to hold a toothpick between the pads of his thumb and index finger
 - D. the middle and distal phalanges of the ring and little finger are flexed at the proximal interphalangeal joint (PIP's) and the distal interphalangeal joint (DIP's)
 - E. all of the above (A-D)
1202. Select the correct statement(s): E
- A. the perforating arteries of the hand connect the deep palmar arch to the dorsal metacarpal arteries on the dorsum of the hand
 - B. the flexor carpi ulnaris and extensor carpi ulnaris are the primary muscles responsible for ulnar flexion of the hand at the wrist
 - C. the palmar metacarpal arteries connect the deep palmar arch to the vessels of the superficial palmar arch
 - D. abduction of the hand at the wrist is primarily carried out by the extensor carpi radialis longus and brevis and flexor carpi radialis
 - E. all of the above (A-D)
- Damage to the musculocutaneous nerve in the axilla will cause: B
- A. loss of arm abduction to the horizontal position at the shoulder joint
 - B. loss of cutaneous innervation extending from the lateral aspect of the forearm to the wrist
 - C. loss of supination of the forearm and hand
 - D. complete loss of flexion of the forearm at the elbow joint due to loss of innervation to the brachialis and biceps brachii
 - E. all of the above (A-D)
- In an amputation of the middle finger at the metacarpophalangeal joint, select the structure least likely to be severed: B
- A. tendon of the extensor digitorum and beginning of dorsal expansion to the middle finger
 - B. the second lumbrical, second palmar interosseous, and third dorsal interosseous, tendons of flexor digitorum superficialis and flexor digitorum profundus
 - C. the second lumbrical, second and third dorsal interossei (or their tendons), tendons of flexor digitorum superficialis and flexor digitorum profundus
 - D. two palmar proper digital and two dorsal proper digital arteries and nerves

1187. Which one of the following terms does not describe a common motion of the shoulder girdle: A
- A. pronation
 - B. retraction
 - C. elevation
 - D. depression
 - E. forward rotation
188. A nervous, new phlebotomist (blood drawer) has inserted the needle too deeply while trying to draw blood from the median cubital vein. Bright red arterial blood enters the syringe. Select the incorrect statement: D
- A. the needle has entered the brachial artery
 - B. the needle in the artery lies lateral to the median nerve
 - C. the needle has pierced the bicipital aponeurosis
 - D. the posterior brachial cutaneous nerve has also been injured
1189. Select the correct statement(s): D
- A. the lateral antebrachial cutaneous nerve accompanies the basilic vein in the region below the elbow joint
 - B. the antebrachial fascia must be pierced during normal blood drawing from the median cubital vein
 - C. the musculocutaneous nerve gives sensory innervation to the medial aspect of the forearm
 - D. the basilic vein dives deep about the midarm to drain into the deep veins of the upper extremity
 - E. all of the above (A-D)
- Select the incorrect statement: D
- A. the triceps brachii muscle crosses both the elbow and shoulder joints
 - B. the biceps brachii muscle is capable of acting on three joints of the upper extremity
 - C. the principal flexors of the forearm at the elbow joint are the brachialis and the biceps brachii
 - D. the principal extensor of the forearm at the elbow joint is the coracobrachialis
1191. Select the correct statement(s): E
- A. a puncture wound to the middle of the palm by a rusty nail which penetrates into the midpalmar compartment must pass through the palmar aponeurosis
 - B. the extensor digitorum must relax to allow the lumbricals (and interossei) to flex the fingers at the metacarpal-phalangeal (M-P) joints
 - C. the lumbricals and superficial palmar arch are located in the midpalmar compartment
 - D. the palmaris brevis is an intrinsic hand muscle which is not found within one of the five, palmar compartments of the hand
 - E. all of the above (A-D)
- Select the correct statement(s): A
- A. severing the median nerve proximal to the cubital fossa will result in complete loss of pronation of hands (Do not consider brachioradialis)
 - B. severing the deep radial nerve during elbow surgery will result in complete loss of supination of the hand (Do not consider brachioradialis)
 - C. the superficial branch of the radial nerve is accompanied by the posterior interosseous artery
 - D. The cephalic vein is accompanied by branches of the medial antebrachial cutaneous nerve in the forearm
 - E. all of the above (A-D)

The clavipectoral fascia:

A

- A. surrounds the pectoralis minor muscle
- B. is pierced by the axillary artery
- C. forms the axillary portion of the breast
- D. forms the axillary sheath
- E. contributes the suspensory ligaments of the breast

Pick the best association. Pectoralis major:

C

- A. medial rotation
- B. adduction
- C. both A and B
- D. neither A nor B

Pick the best association. Thoraco-acromial trunk:

A

- A. axillary artery
- B. subclavian artery
- C. both A and B
- D. neither A nor B

Branches of the third part of the axillary artery include:

A

- A. subscapular
- B. thoracoacromial trunk
- C. thyrocervical trunk
- D. lateral thoracic
- E. all of the above (A-D)

1177 Which of the following structures passes superficial to the flexor retinaculum of the wrist:

B

- A. radial artery
- B. tendon of palmaris longus
- C. median nerve
- D. tendon of flexor carpi radialis
- E. all of the above (A-D)

1178. The shoulder joint is weakest and least stable:

C

- A. anteriorly
- B. posteriorly
- C. inferiorly
- D. superiorly
- E. all of the above (A-D)

1179. A biker was thrown over the handlebars of his motorcycle and landed on his right shoulder and right side of his head. He excessively displaced his head toward the left and depressed his right shoulder. Select the correct statement(s) about the results of the fall:

E

- A. he will not be able to comb his hair with his right hand due to loss of innervation to his deltoid and supraspinatus muscles
- B. he will not be able to feel his dates run their hands along the lateral aspect of his forearm
- C. his right forearm and hand will be pronated due to loss of innervation to the muscles supplied by the musculocutaneous nerve
- D. he now appears to be a waiter looking for a tip. Have no fear! He will recover with the proper physical therapy!
- E. all of the above (A-D)

1156. Dislocation is more likely to occur at the sternoclavicular joint than at the acromioclavicular joint: B
A. true
B. false
157. The extensor indicis, extensor digitorum and extensor digiti minimi would all be used to extend the fingers at the metacarpalphalangeal joints: A
A. true
B. false
1158. Following reduction of an inferior dislocation of the right shoulder joint in a 17-year old male, you discover that abduction in this joint is very weak. Based on your anatomical knowledge of this region, which nerve was most likely traumatized during the dislocation: B
A. suprascapular
B. axillary
C. upper and lower subscapular
D. both A and B
E. all of the above (A-C)
1159. Which one of the following components of the brachial plexus would truly be located within the borders of the axilla: C
A. roots
B. trunks
C. cords
D. divisions
E. origin of the subclavian nerve
- A "Colles" fracture usually involves the distal ulna: B
A. true
B. false
- Which one of the following statements is correct: D
A. the scapular notch transmits the subscapular nerve but not the artery of the same name
B. the triangular interval transmits a small branch of the radial nerve
C. the axillary artery has its origin at the medial border of the first rib
D. pectoralis minor is innervated by the medial pectoral nerve
E. subscapularis passes anterior to the humerus to insert upon the greater tuberosity
1162. Which one of the following statements is not correct: B
A. the point of insertion for the deltoid is the deltoid tuberosity
B. the intertubercular sulcus provides a pathway for the tendon of the long head of triceps brachii
C. the cephalic vein passes through the clavipectoral fascia
D. the lateral pectoral nerve passes to pectoralis major medial to pectoralis minor
E. pectoralis minor can assist breathing under special circumstances
- The radial artery is palpable in between the tendons of: D
A. extensor pollicis brevis and abductor pollicis brevis
B. abductor pollicis longus and abductor pollicis brevis
C. abductor pollicis longus and flexor pollicis brevis
D. extensor pollicis brevis and extensor pollicis longus
E. extensor carpi radialis brevis and extensor carpi radialis longus

1140. The digital fibrous sheath around the tendons of the flexor digitorum superficialis and profundus act as "pulleys" and increase the efficiency of the muscles by holding the tendons close to the bone: A

A. true
B. false

At the wrist, the radial artery lies lateral to the tendons of the abductor pollicis longus and extensor pollicis brevis: A

A. true
B. false

1142. The primary flexors of the distal interphalangeal joints are the flexor digitorum superficialis and profundus: B

A. true
B. false

The extensor pollicis longus tendon has its own synovial sheath as it passes beneath the extensor retinaculum: A

A. true
B. false

An example of a precision grip is gripping a pencil as you write: A

A. true
B. false

The superficial palmar arch lies between the palmar aponeurosis and the tendons of the flexor digitorum superficialis: A

A. true
B. false

The vincula longus and brevis going to the tendons in the digits are the sole blood source for this area of the tendons: A

A. true
B. false

1147. The most important functional movements of the hand occur between the thumb and the index and middle fingers: A

A. true
B. false

Of the following descriptive remarks concerning the posterior cord of the brachial plexus, which one is correct: E

- A. it is composed of all the anterior divisions
- B. the nerves to teres major, teres minor, deltoid and rhomboideus major in the shoulder region arise from the posterior cord
- C. the posterior cord makes a major contribution to the median nerve
- D. cutting this cord would greatly impair lateral rotation of the arm at the shoulder
- E. the name of the nerve derived from the posterior cord which innervates latissimus dorsi is the thoracodorsal nerve

The triceps brachii is primarily an extender of the forearm and a weak extensor of the arm: A

- A. true
- B. false

The tendon of the long head of the biceps brachii lies within the bicipital groove (intertubercular groove) covered by the transverse humeral ligament: A

- A. true
- B. false

The bicipital aponeurosis separates the median nerve and the brachial artery as they run medial to the tendon of the biceps brachii in the cubital fossa: B

- A. true
- B. false

In making a fist during a fight, it is necessary for the flexor carpi ulnaris and radialis, the extensor carpi ulnaris and the extensor carpi radialis longus and brevis to synergistically tighten to stabilize the wrist: A

- A. true
- B. false

Motor impulses to skeletal muscle course through the musculocutaneous nerve distal to the cubital fossa: B

- A. true
- B. false

One route of collateral circulation around the elbow is the radial collateral artery off the profunda brachii artery anastomosing with the radial recurrent artery off the radial artery: A

- A. true
- B. false

Attempts are often made to reconstitute a normal relationship of bones after injuries. It is, therefore, important to know the usual relationship. At the wrist, the following one is true for the relationship between the radius and the ulna: A

- A. the radius extends about one centimeter more distally than the ulna
- B. the ulna extends about one centimeter more distally than the radius
- C. the relationship is normally variable
- D. the styloid processes of the ulna and radius are ordinarily at the same level as far as their distal extension is concerned
- E. none of the above (A-D)

Structures found within the "anatomical snuffbox" include: A

- A. radial artery
- B. radial nerve
- C. both A and B
- D. neither A nor B

The axillary sheath fuses and merges with the lateral intermuscular septum of the arm: B

- A. true
- B. false

103. The ulna: E
- A. is subject to parry fractures
 - B. has its major growth in its distal portion
 - C. is on the little finger side of the hand
 - D. both A and C
 - E. all of the above (A-C)
1104. The nerve supply to pectoralis major branches off from the: A
- A. lateral cord of the brachial plexus
 - B. posterior cord of the brachial plexus
 - C. both A and B
 - D. neither A nor B
1105. All of the following are found in the axilla except: D
- A. apical group of lymph nodes
 - B. third part of the axillary artery
 - C. upper trunk of brachial plexus
 - D. dorsal scapular nerve
 - E. medial brachial cutaneous nerve
1106. Which of the following muscles are the principal ones for flexion of the hand at the wrist: D
- A. pronator teres and brachioradialis
 - B. brachioradialis and palmaris longus
 - C. flexor pollicis longus and flexor digitorum profundus
 - D. flexor carpi radialis and flexor carpi ulnaris
 - E. all of the above (A-D)
1107. In a fracture of the clavicle, the lateral fragment is displaced downward. This is due to the pull of the: A
- A. upper limb
 - B. trapezius
 - C. pectoralis major
 - D. sternocleidomastoid
 - E. subclavius
- Sensory innervation over the acromial process branches off from the: D
- A. lateral cord of the brachial plexus
 - B. posterior cord of the brachial plexus
 - C. both A and B
 - D. neither A nor B
1109. Medial rotation of humerus is produced by which muscle: B
- A. teres minor
 - B. teres major
 - C. both A and B
 - D. neither A nor B
- All of the following nerves are derived from a cord of the brachial plexus except: C
- A. thoracodorsal (middle subscapular)
 - B. axillary
 - C. long thoracic
 - D. lateral pectoral
 - E. medial pectoral

1087. Tendons of muscles contributing to the "rotator cuff" do not include: E
- A. subscapularis
 - B. supraspinatus
 - C. infraspinatus
 - D. teres minor
 - E. teres major
1088. The tendon of which muscle is a good guide to finding the ulnar artery at the wrist: B
- A. palmaris longus
 - B. flexor carpi ulnaris
 - C. flexor carpi radialis
 - D. flexor carpi radialis longus
1089. Each cord of the brachial plexus: D
- A. contains nerve fibers from C5 to T1 levels of the spinal cord
 - B. is formed from the junction of one anterior and one posterior division
 - C. gives a branch to the median nerve
 - D. lies within the axilla
 - E. gives rise to a subscapular nerve i.e. upper, middle, and lower
- Which of the following is not derived from medial cord: C
- A. medial brachial cutaneous
 - B. medial antebrachial cutaneous
 - C. musculocutaneous
 - D. ulnar
 - E. medial pectoral
1091. Which of the following does not receive at least partial blood supply from branches of the axillary artery: E
- A. skin over deltopectoral triangle
 - B. lateral breast
 - C. latissimus dorsi muscle
 - D. serratus anterior muscle
 - E. thyroid gland
1092. All phalanges and the first metacarpal usually have only one distal epiphysis: B
- A. true
 - B. false
1093. The following features are located laterally on the humerus: D
- A. greater tubercle
 - B. radial groove
 - C. trochlear process
 - D. both A and B
 - E. all of the above (A-C)
- The most frequently fractured carpal bone in the hand is the: D
- A. trapezium
 - B. pisiform
 - C. lunate
 - D. scaphoid
 - E. capitate

1072. Damage to which nerve would result in an inability to flex the arm: D
- A. radial
 - B. ulnar
 - C. median
 - D. musculocutaneous
 - E. axillary
1073. Damage to which nerve would result in an inability to oppose the thumb: C
- A. radial
 - B. ulnar
 - C. median
 - D. musculocutaneous
 - E. axillary
1074. Damage to which nerve would result in an inability to adduct and abduct the fingers: B
- A. radial
 - B. ulnar
 - C. median
 - D. musculocutaneous
 - E. axillary
1075. Muscles which insert on the floor or crest of the intertubercular sulcus of the humerus include: E
- A. latissimus dorsi
 - B. pectoralis major
 - C. teres major
 - D. both A and B
 - E. all of the above (A-C)
1076. The anterior interosseus nerve is a branch of the: D
- A. ulnar nerve
 - B. radial nerve
 - C. musculocutaneous nerve
 - D. median nerve
 - E. common interosseous nerve
1077. In an industrial accident a man had a sharp implement cut down to the base of the metacarpals of the first interosseous space, cutting the first dorsal interosseus and the adductor pollicis muscles. There is accompanying spurting of bright red blood. What vessel has most likely been severed: E
- A. second common digital artery
 - B. cephalic vein
 - C. ulnar artery
 - D. princeps pollicis artery
 - E. none of the above (A-D)
1078. Which structure would not be associated with the cubital fossa: C
- A. brachial artery
 - B. biceps brachii tendon
 - C. coracobrachialis
 - D. median nerve
 - E. brachioradialis

1056. Which of the following muscles flex the forearm: A
- A. brachialis
 - B. coracobrachialis
 - C. both A and B
 - D. neither A nor B
1057. Muscles in the anterior (flexor forearm) which are supplied by the ulnar nerve include: B
- A. flexor digitorum superficialis
 - B. flexor carpi ulnaris
 - C. both A and B
 - D. neither A nor B
1058. Klumpke's palsy can be caused by excess abduction of the arm: A
- A. true
 - B. false
1059. Structures passing deep to or within the flexor retinaculum include all but: A
- A. tendon of palmaris longus
 - B. tendon of flexor digitorum superficialis
 - C. median nerve
 - D. tendon of flexor digitorum profundus
 - E. all of the above pass within the flexor retinaculum
1060. One of the Anatomy professors tutored first-year medical students too late one night and fell asleep with his/her right upper limb hanging over the back of a chair. The next morning he/she exhibited wrist drop, numbness along the posterior limb and poor hand grip. which nerve was temporarily injured: B
- A. ulnar
 - B. radial
 - C. musculocutaneous
 - D. median
 - E. accessory
1061. Which of the following statements concerning the brachial plexus is not true: C
- A. ventral rami of C5 and C6 unite to form the upper trunk
 - B. the subscapular nerve originate from the posterior cord
 - C. all anterior divisions unite to form the posterior cord
 - D. ventral rami of C8 and T1 unite to form the lower trunk
 - E. each trunk divides into an anterior and posterior division
- The lateral epicondyle of the humerus provides an origin for: E
- A. extensor carpi ulnaris
 - B. extensor digitorum
 - C. extensor carpi radialis brevis
 - D. both A and B
 - E. all of the above (A-C)
- Extrinsic thumb muscles, that are sometimes referred to as "outcropping" muscles, include: E
- A. extensor pollicis brevis
 - B. extensor pollicis longus
 - C. abductor pollicis longus
 - D. both A and B
 - E. all of the above (A-C)

Primary curvatures of the vertebral column include:

B

- A. lumbar
- B. sacral
- C. both A and B
- D. neither A nor B

A typical cervical vertebra has everything except:

A

- A. an odontoid process
- B. a foramen transversarium in each transverse process
- C. small, bifid spines
- D. a large triangular vertebral forame

1030. Which nerve innervates the rhomboid major muscle: C
- A. dorsal rami of spinal nerves
 - B. spinal accessory nerve
 - C. dorsal scapular nerve
 - D. C₃ + C₄
 - E. suprascapular nerve
1031. Which nerve innervates the sternocleidomastoid muscle: B
- A. dorsal rami of spinal nerves
 - B. spinal accessory nerve
 - C. dorsal scapular nerve
 - D. C₃ + C₄
 - E. suprascapular nerve
- All of the following are true about thoracolumbar fascia except: D
- A. it extends laterally from lumbar and sacral spines
 - B. it forms the roof of the lumbar triangle
 - C. the intertransverse ligament divides to form the anterior and middle layer of thoracolumbar fascia
 - D. the posterior lamella divides and encloses the quadratus lumborum muscle
 - E. the posterior lamella gives origin to the latissimus dorsi muscle
- The trapezius muscle: D
- A. receives its motor supply from the accessory nerve
 - B. inserts in part upon the humerus
 - C. rotates the scapula
 - D. both A and C
 - E. all of the above (A-C)
- The intervertebral discs: C
- A. are classified as symphyses
 - B. vary in thickness in different regions of the vertebral column
 - C. both A and B
 - D. neither A nor B
1035. Which nerve innervates the levator scapula muscle: D
- A. dorsal rami of spinal nerves
 - B. spinal accessory nerve
 - C. dorsal scapular nerve
 - D. C₃ + C₄
 - E. suprascapular nerve
1036. An abnormal posterior curvature of the vertebral column in the thoracic region, commonly observed in elderly women, is known as: D
- A. scoliosis
 - B. osteoporosis
 - C. lordosis
 - D. kyphosis
 - E. tuberculosis
1037. The muscles of the deep, intrinsic group, the "true" back muscles are: E
- A. concerned with the maintenance of upright posture
 - B. supplied by branches of the dorsal primary rami
 - C. control the movements of the vertebral column
 - D. both A and C
 - E. all of the above (A-D)

1014. The spinal cord usually terminates opposite the intervertebral disc between which vertebrae in the adult:

A

- A. T₁₂ + L₁
- B. L₃ + L₄
- C. L₄ + L₅
- D. S₁ + S₂
- E. none of the above (A-D)

Which nerve innervates the splenius cervicis muscle:

A

- A. dorsal rami of spinal nerves
- B. spinal accessory nerve
- C. dorsal scapular nerve
- D. C₃ + C₄
- E. suprascapular nerve

The ligamentum flavum:

B

- A. bridges the skull and atlas
- B. runs between 2 adjacent laminae
- C. runs between 2 spinous processes
- D. runs between 2 transverse processes
- E. none of the above (A-D)

The posterior intervertebral joints are classified as cartilaginous joints:

B

- A. true
- B. false

The odontoid process is:

E

- A. the "captured" body of C₁
- B. part of the axis vertebra
- C. also known as the dens
- D. both A and C
- E. all of the above (A-C)

When an adult is standing erect, the conus medullaris will most usually be located at about the level of:

B

- A. T₁ vertebra
- B. L₁ vertebra
- C. L₃ vertebra
- D. S₅ vertebra
- E. none of the above (A-D)

The superficial muscles of the back are supplied by:

D

- A. dorsal rami of spinal nerves
- B. ventral roots of spinal nerves
- C. both A and B
- D. neither A nor B

1021. The spinal cord proper extends downward from the foramen magnum superiorly to the inferior border of the first lumbar vertebra where it terminates as the:

D

- A. filum terminale
- B. coccygeal ligament
- C. cauda equina
- D. conus medullaris
- E. dentate ligament

The first coccygeal segment of the spinal cord would be located approximately at vertebral level: C

- A. C1
- B. T1
- C. L2
- D. S2
- E. coccygeal 1

A heart-shaped vertebral body is seen in which vertebrae: B

- A. cervical
- B. thoracic
- C. lumbar
- D. sacral
- E. coccygeal

Which of the following ligaments would prevent hyperextension of the vertebral column: A

- A. anterior longitudinal ligament
- B. posterior longitudinal ligament
- C. ligamentum flava
- D. iliofemoral ligament
- E. interspinous ligament

"Hunchback" refers to: B

- A. lordosis
- B. an abnormal thoracic curvature
- C. an abnormal lumbar curvature
- D. both A and B
- E. both A and C

The spinal accessory nerve (cranial nerve XI) supplies the: C

- A. trapezius
- B. sternocleidomastoid
- C. both A and B
- D. neither A nor B

The suboccipital triangle: E

- A. is bounded by obliquus capitis inferior, obliquus capitis superior, and rectus capitis major
- B. contains the vertebral artery
- C. contains the suboccipital nerve
- D. both A and B
- E. all of the above (A-C)

1003. The vertebral column includes: E

- A. symphysial joints
- B. synovial joints
- C. syndesmoses
- D. both A and B
- E. all of the above (A-C)

Foramina transversaria: A

- A. transmit the vertebral artery
- B. transmit spinal nerves
- C. are located in the transverse processes of all cervical vertebrae
- D. are located within the pedicles

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GROSS ANATOMY:
PELVIS & PERINEUM



Which of the following is not correct:

C

- A. the layers making up the anterior abdominal wall become continuous with the layers surrounding the penis and scrotum in the perineum
- B. the dorsal arteries and nerves of the penis are paired, but the superficial and deep dorsal veins of the penis are single midline structures
- C. the urethral crest and seminal colliculus are found in the membranous portion of the male urethra
- D. the bulb of the penis continues into the body as the corpus spongiosum and terminates as the glans penis

975. Bimanual pelvic examination (one or two fingers of one hand in the vagina and other hand pressing on anterior abdominal wall) of a 49 kg nulliparous woman should allow you to demonstrate and examine all of the following except the:

C

- A. uterine fundus
- B. ovary
- C. lower pole of kidney
- D. rectouterine pouch (of Douglas)

976. Which of the following is/are correct:

B

- A. the round ligament of the uterus remains outside the layers of the broad ligament throughout the course
- B. major supports of the uterus include the levator ani muscles and condensations of pelvic fascia which condense to form ligaments
- C. both A and B
- D. neither A nor B

Which of the following is/are correct:

B

- A. the suspensory ligament of the ovary extends from the upper end of the lateral wall of the uterus to the medial portion of the ovary
- B. the uterine artery ascends within the broad ligament along the lateral margin of the uterus to anastomose with the ovarian artery
- C. the most distal portion of the uterine tube is referred to as the ampulla
- D. the artery supplying the ovary is the ovarian artery, a branch of the internal iliac
- E. all of the above (A-D)

Which is/are located in the deep perineal pouch:

C

- A. greater vestibular gland
- B. sphincter vaginae muscle
- C. dorsal nerve of the clitoris (penis)
- D. pudendal nerve
- E. all of the above (A-D)

The uterine cervix:

E

- A. comprises approximately one-third of the uterus
- B. joins the remainder of the uterus at the internal os
- C. joins the vagina at a near-right angle normally
- D. is the best stabilized portion of the uterus
- E. all of the above (A-D)

The pelvic diaphragm:

E

- A. forms the inferior boundary of the pelvic cavity
- B. can be divided into levator ani and coccygeus muscles
- C. exhibits midline openings for the rectum and urethra in the male
- D. is attached to the pubis, lateral pelvic wall (arcus tendineus) and ischial spine
- E. all of the above (A-D)

957. The posterior femoral cutaneous nerve passes through the: A
- A. greater sciatic foramen
 - B. lesser sciatic foramen
 - C. obturator foramen
 - D. both A and B
 - E. all of the above (A-C)
958. Concerning the levator ani muscle: C
- A. its puborectal portion can be palpated during rectal examination
 - B. together with the ischiococcygeus muscle, it forms the pelvic diaphragm
 - C. both A and B
 - D. neither A nor B
959. Concerning the uterus: E
- A. it is normally anteverted and anteflexed
 - B. it forms from the fused lower ends of the paramesonephric ducts
 - C. the fundus lies above the entrance of the uterine tubes
 - D. the internal os is the junction of the uterine cavity with the cervical canal
 - E. all of the above (A-D)
- Select the correct statement(s): B
- A. the trabecula within the corpora cavernosa become engorged with blood during erection
 - B. the principal arteries for filling the erectile tissue are the deep arteries
 - C. both A and B
 - D. neither A nor B
961. The obturator internus tendon passes through the: B
- A. greater sciatic foramen
 - B. lesser sciatic foramen
 - C. obturator foramen
 - D. both A and B
 - E. all of the above (A-C)
- Which of the following is/are true regarding the perineal pouches: D
- A. the bulbourethral glands lie within the superficial pouch
 - B. the deep pouch in the male contains the prostatic portion of the urethra
 - C. both A and B
 - D. neither A nor B
963. Which of the following is/are found between the two peritoneal layers forming the broad ligament: A
- A. epoophoron
 - B. urethra
 - C. both A and B
 - D. neither A nor B
- The following statements are true about the levator ani muscle except it: D
- A. is important in fecal continence
 - B. does not receive a contribution from the ischiococcygeus muscle
 - C. takes a portion of its origin from the obturator internus fascia
 - D. is pierced by the anus in both sexes as well as the vagina in females
- The vaginal cervix: E
- A. enters the anterior wall of the vagina
 - B. is surrounded by the vaginal fornices
 - C. joins the vagina at a near-right angle
 - D. contains the external os
 - E. all of the above (A-D)

941. Double folds of peritoneum include all of the following except: C
- A. mesosalpinx
 - B. mesovarium
 - C. suspensory ligament of the ovary
 - D. mesometrium
 - E. broad ligament
942. The peroneal artery is a: C
- A. direct branch of femoral artery
 - B. direct branch of popliteal artery
 - C. direct branch of posterior tibial artery
 - D. all of the above (A-C)
 - E. none of the above (A-D)
943. The medial plantar artery is a: C
- A. direct branch of femoral artery
 - B. direct branch of popliteal artery
 - C. direct branch of posterior tibial artery
 - D. all of the above (A-C)
 - E. none of the above (A-D)
944. The inferior gluteal artery is a: E
- A. direct branch of femoral artery
 - B. direct branch of popliteal artery
 - C. direct branch of posterior tibial artery
 - D. all of the above (A-C)
 - E. none of the above (A-D)
945. The anterior tibial artery is a: B
- A. direct branch of femoral artery
 - B. direct branch of popliteal artery
 - C. direct branch of posterior tibial artery
 - D. all of the above (A-C)
 - E. none of the above (A-D)
946. In the male, an intact sacral spinal cord is thought to be important in the: C
- A. production of an erection
 - B. ejaculation of seminal products
 - C. both A and B
 - D. neither A nor B
947. The internal pudendal artery passes through the: D
- A. greater sciatic foramen
 - B. lesser sciatic foramen
 - C. obturator foramen
 - D. both A and B
 - E. all of the above (A-C)
948. Which of the following is/are correct: D
- A. the paraurethral glands in the female are homologous to the seminal vesicles in the male
 - B. paired sinovaginal bulbs grow up from the urogenital sinus to form the cervix
 - C. the epithelium of the vagina is derived from mesoderm of the urogenital sinus
 - D. the greater vestibular glands (of Bartholin) in the female are homologous to the bulbourethral glands in the male
 - E. all of the above (A-D)

Which of the following statements about the inferior pelvic aperture is true:

D

- A. it is bordered by the pelvic brim
- B. it is closed by the urogenital diaphragm
- C. it denotes the entry into the true pelvis
- D. it is formed anteriorly by the pubic arch and posteriorly by the sacrotuberous ligaments
- E. the conjugate diameter is a measure of its anterior-posterior diameter

The mature female pelvis is characterized by:

E

- A. a wide sub-pubic angle
- B. smaller acetabula
- C. a wide sciatic notch
- D. both A and C
- E. all of the above (A-C)

Compared with the mature male sacrum, the female sacrum:

E

- A. is wider
- B. is shorter
- C. has a smaller lumbosacral articulation
- D. both A and B
- E. all of the above (A-C)

928. The "greater" pelvis:

D

- A. includes the bladder in infancy
- B. is the "false" pelvis
- C. is below the pelvic inlet
- D. both A and B
- E. all of the above (A-C)

929. Which does not contribute to the walls of the ischiorectal fossa:

C

- A. iliococcygeus muscle
- B. pubococcygeus muscle
- C. bulbospongiosus muscle
- D. obturator internus muscle
- E. coccygeus muscle

All of the following are in the deep perineal pouch except the:

A

- A. levator prostate muscle
- B. bulbourethral gland
- C. membranous urethra
- D. deep transverse perineus muscle
- E. dorsal artery of the penis/clitoris

931. Which of the following is in the vestibule of the female perineum:

A

- A. ducts of greater vestibular gland
- B. perineal body
- C. crus of clitoris
- D. labia majora

All of the following are true of Colles' fascia except:

B

- A. it is superficial fascia
- B. it is a thick, fatty layer
- C. it is continuous with Scarpa's fascia
- D. it forms the inferior limit of the superficial perineal pouch
- E. it fuses with the posterior margin of the urogenital diaphragm

909. Thrombosis of the deep penile arteries would most likely lead to: **B**
- A. necrosis of the penis
 - B. inability to produce an erection
 - C. prolongation of an erection, once obtained
 - D. both A and B
 - E. all of the above (A-C)
910. Which of the following is/are correct: **D**
- A. the urogenital diaphragm forms a muscular sling just superior to the prostate
 - B. seminal vesicles are two multilobulated organs in the male which produce citric acid and acid phosphatase
 - C. both A and B
 - D. neither A nor B
911. All of the following structures open into the prostatic urethra except: **D**
- A. ejaculatory ducts
 - B. prostatic ducts
 - C. bladder
 - D. bulbourethral glands
 - E. prostatic utricle
- Which of the following structures cannot be palpated via the vagina: **C**
- A. cervix
 - B. rectouterine pouch
 - C. vesicouterine pouch
 - D. rectum
 - E. ureters
913. The ejaculatory duct: **C**
- A. enters the prostatic urethra on the seminal colliculus
 - B. traverses the substance of the prostate gland
 - C. both A and B
 - D. neither A nor B
914. The striated muscular component which is a portion of the pelvic diaphragm whose bundles make a loop just behind the junction between the rectum and anus is the: **D**
- A. sphincter ani internus muscle
 - B. corrugation cutis ani muscle
 - C. sphincter externus muscle
 - D. puborectalis muscle
 - E. rectococcygeus muscle
- The inferior hypogastric plexus may receive inputs from all of the following except: **B**
- A. sacral splanchnics
 - B. vagus
 - C. pelvic splanchnics
 - D. superior hypogastric plexus
- The pudendal nerve, as it lies in the pudendal canal, contains nerve fibers: **D**
- A. transmitting cutaneous sensation from the penis to the CNS
 - B. that are lower motor neuron axons to the bulbospongiosus and ischiocavernosus muscles
 - C. which are the primary source of autonomic innervation to the penis
 - D. both A and B
 - E. all of the above (A-C)

893. Which structure is not in or does not pass through the deep perineal pouch of the urogenital triangle: **B**
- A. urethra
 - B. root of the clitoris
 - C. deep transverse perineal muscle
 - D. internal pudendal artery
 - E. vagina
894. Which of the following is not a characteristic of the bony female pelvis: **B**
- A. inlet is transversely oval
 - B. long and narrow cavity
 - C. round and wide subpubic angle
 - D. everted ischial tuberosities
- Select the one correct answer regarding the anal triangle: **C**
- A. the central tendon of the perineum (perineal body) lies posterior to the anus
 - B. the external anal sphincter corresponds to the inner circular layer of muscle seen throughout the digestive tract
 - C. the medial wall of the ischiorectal fossa is the pelvic diaphragm
 - D. the sensory supply to the anal triangle is from the pelvic plexus
896. Which structure is not a boundary of the perineum: **A**
- A. sacrospinous ligament
 - B. pubic symphysis
 - C. sacrotuberous ligament
 - D. ischial tuberosity
 - E. coccyx
897. In the male, an intact sacral spinal cord is thought to be important in the: **D**
- A. emission of seminal products
 - B. resolution of an erection
 - C. both A and B
 - D. neither A nor B
- Which of the following statements is true: **D**
- A. the internal pudendal artery is embedded in a fascial canal within the urogenital triangle
 - B. the pudendal nerve is a branch of the lumbar plexus
 - C. the posterior scrotal (labial) nerves are branches of the inferior rectal nerve
 - D. the pudendal nerve enters the perineum (from the gluteal region) by passing through the lesser sciatic foramen
899. Tactile innervation of the clitoris is mediated via the: **B**
- A. pelvic splanchnics
 - B. sacral dorsal roots (S₂ - S₄)
 - C. both A and B
 - D. neither A nor B
900. The posterior reflection of the peritoneum that represents the lowest portion of the pelvic cavity in the female is called the: **C**
- A. uterovesical pouch
 - B. broad ligament
 - C. rectouterine pouch
 - D. round ligament

The branches of the pudendal nerve are located in the:

E

- A. ischiorectal fossa
- B. deep perineal pouch
- C. superficial perineal pouch
- D. both B and C
- E. all of the above (A-C)

Which of the following is/are true regarding the perineal pouches:

C

- A. the major components of the superficial pouch in the female are the root of the clitoris and the muscles that cover the root
- B. the superficial pouch is bounded by the membranous layer of superficial fascia below and the inferior fascia of the urogenital diaphragm above
- C. both A and B
- D. neither A nor B

Termination of internal pudendal artery occurs in the:

B

- A. ischiorectal fossa
- B. deep perineal pouch
- C. superficial perineal pouch
- D. both B and C
- E. all of the above (A-C)

Primary support for the uterus is the:

D

- A. broad ligament
- B. ovarian ligament
- C. round ligament of the uterus
- D. transverse cervical (cardinal) ligament
- E. more than one of the above

881 A double peritoneal fold is characteristic of the:

A

- A. broad ligament
- B. ovarian ligament
- C. round ligament of the uterus
- D. transverse cervical (cardinal) ligament
- E. more than one of the above

Which of the following is an embryonic vestige:

E

- A. broad ligament
- B. ovarian ligament
- C. round ligament of the uterus
- D. transverse cervical (cardinal) ligament
- E. more than one of the above

883. Which of the following is/are found between the two peritoneal layers forming the broad ligament:

B

- A. superior hypogastric plexus
- B. branches of the ovarian artery
- C. both A and B
- D. neither A nor B

The control of smooth muscle tone in the blood vessels of erectile tissue that allows erection to occur is primarily a function of the:

B

- A. cranial parasympathetic centers
- B. sacral parasympathetic centers
- C. upper thoracic sympathetic centers
- D. lower thoracic and lumbar sympathetic centers

- 861 The bulb of the vestibule is located in the: **B**
- A. deep perineal pouch
 - B. superficial perineal pouch
 - C. ischiorectal fossa
 - D. none of the above (A-C)
- The bulb of the penis is located in the: **B**
- A. deep perineal pouch
 - B. superficial perineal pouch
 - C. ischiorectal fossa
 - D. none of the above (A-C)
- 863 Retroversion refers to the angle between the: **C**
- A. fundus and body of the uterus
 - B. uterine cavity and vaginal cavity
 - C. cervical canal and vaginal cavity
 - D. uterine cavity and cervical canal
 - E. none of the above (A-D)
- Primary support for the uterus is derived from the: **A**
- A. lateral (transverse) cervical ligaments
 - B. pubocervical ligaments
 - C. sacrocervical ligaments
 - D. round ligament of the uterus
 - E. broad ligament
865. When viewed from behind, the order from lateral to medial of structures found on the base of the urinary bladder is: **D**
- A. seminal vesicle $\Rightarrow$ ureter $\Rightarrow$ ampulla of vas deferens
 - B. ureter $\Rightarrow$ ampulla of vas deferens $\Rightarrow$ seminal vesicle
 - C. ampulla of vas deferens $\Rightarrow$ ureter $\Rightarrow$ seminal vesicle
 - D. ureter $\Rightarrow$ seminal vesicle $\Rightarrow$ ampulla of vas deferens
 - E. ampulla of vas deferens $\Rightarrow$ seminal vesicle $\Rightarrow$ ureter
- Which is not considered a usual branch of the anterior division of the internal iliac artery: **B**
- A. internal pudendal artery
 - B. iliolumbar artery
 - C. obturator artery
 - D. umbilical artery
 - E. middle rectal artery
- Which statement is true regarding the nerves and arteries to the pelvis: **D**
- A. the pelvic splanchnic nerves (S₂, 3, 4) constitute the sacral part of the sympathetic system
 - B. the hypogastric nerves contribute parasympathetic components to the pelvic plexus
 - C. the superior and inferior gluteal arteries are branches of the external iliac artery while the middle rectal artery is a branch of the internal iliac artery
 - D. the pudendal nerve is from the sacral plexus (S₂, 3, 4) and has a short course in the gluteal region
868. Which one of the following is true regarding the boundaries of the pelvis: **C**
- A. the false or greater pelvis lies just below the pelvic brim
 - B. the true pelvis has a short posterior wall and a much deeper anterior wall
 - C. the true pelvis (pelvis minor) lies between the pelvic brim and the perineum
 - D. a portion of the parietal pelvic fascia overlying the piriformis muscle is condensed to form the tendinous arch

844. The abdominal aorta: A
- A. is located retroperitoneally
 - B. passes through the diaphragm
 - C. both A and B
 - D. neither A nor B

845. The vertebrocostal triangle: D
- A. allows for passage of the inferior vena cava
 - B. is located within the central tendon of the diaphragm
 - C. both A and B
 - D. neither A nor B

- The lumbar splanchnic nerves: D
- A. originate from neurons located in the lumbar sympathetic ganglia
 - B. make synaptic contact in the walls of the abdominal viscera
 - C. both A and B
 - D. neither A nor B

- The pectinate line in the anal canal represents a transition area between all of the following except the: B
- A. vascular supply from the inferior mesenteric and internal iliac arteries
 - B. parasympathetic preganglionic fibers of vagal origin and those from the sacral spinal cord
 - C. visceral afferent and somatic afferents
 - D. mucous membrane lining the GI tract and stratified squamous epithelium of the skin

- The membranous layer of superficial fascia (Scarpa's fascia) of the anterior abdominal wall: B
- A. forms one of the three fascial layers surrounding the spermatic cord
 - B. is attached inferiorly to the inguinal ligament
 - C. both A and B
 - D. neither A nor B

- The transversalis fascia: C
- A. is continuous with the deep fascia of the diaphragm, the parietal pelvic fascia and the thoracolumbar fascia
 - B. forms one of the three fascial layers surrounding spermatic cord
 - C. both A and B
 - D. neither A nor B

- The following is true about a direct inguinal hernia except it: D
- A. occurs within the inguinal triangle
 - B. usually travels between the inner and middle spermatic fascia
 - C. enters the inguinal canal through its posterior wall
 - D. enters the inguinal canal through deep inguinal ring

- 851 The cremaster muscle: E
- A. is innervated by the genital branch of the genitofemoral nerve
 - B. is continuous with the internal abdominis oblique muscle
 - C. forms part of the middle spermatic fascia
 - D. functions to raise the testes
 - E. all of the above (A-D)

- The inguinal ligament: B
- A. passes directly inferior to the femoral canal
 - B. forms the inferior border of the external abdominis oblique muscle
 - C. both A and B
 - D. neither A nor B

828. The thoracic splanchnic nerves: D
- A. form synaptic connections in the aortic plexus (intermesenteric plexus)
 - B. provide parasympathetic input to the viscera of the abdomen
 - C. both A and B
 - D. neither A nor B
829. The following statements are true about an indirect inguinal hernia except it: B
- A. occurs lateral to the inferior epigastric artery
 - B. usually lies between the inner and middle spermatic fascia
 - C. is more common in males than females
 - D. is often associated with a patent processus vaginalis
830. The iliacus muscle: E
- A. arises from the iliac fossa
 - B. receives its motor innervation from L2, 3, and 4
 - C. inserts in common with the psoas major
 - D. serves to flex the thigh
 - E. all of the above (A-D)
831. The ventral mesentery: D
- A. extends from the foregut to the hindgut
 - B. forms the greater omentum
 - C. early in development, contains small intercellular clefts which assist formation of the omental bursa
 - D. forms the falciform ligament and the lesser omentum
 - E. all of the above (A-D)
832. Which of the following is not true regarding the diaphragm: B
- A. it has a peripheral muscular part which inserts into a central tendon
 - B. it is considered one of the muscles of expiration
 - C. it has an extensive origin from areas of the sternum, ribs, vertebrae, and arcuate ligaments
 - D. the motor nerve supply to the entire diaphragm is phrenic (C3, 4, 5)
- Above the arcuate line, the anterior layer of the rectus sheath is composed of aponeurosis of all of the following muscles except the: C
- A. external abdominal oblique muscle
 - B. internal abdominal oblique muscle
 - C. rectus abdominis muscle
 - D. transversus abdominal muscle
834. Which of the following is/are true regarding the nerves of the posterior abdominal wall: A
- A. the lumbar plexus is formed from ventral rami of the upper four lumbar nerves within the psoas major muscle
 - B. the sympathetic trunk enters the abdomen (from the thorax) under the lateral arcuate ligament
 - C. both A and B
 - D. neither A nor B
835. A patient has a perforation of an ulcer located on the posterior wall of the stomach. Which of the following structures least likely to be damaged: C
- A. spleen
 - B. pancreas
 - C. left suprarenal gland
 - D. liver

Scarpa's (membranous) fascia is:

B

- A. continuous with the external spermatic fascia
- B. attached inferiorly to the iliac crest, inguinal ligament and ischiopubic ramus
- C. both A and B
- D. neither A nor B

812. The diaphragm:

A

- A. is attached to the bodies of the lumbar vertebrae via the crura
- B. receives its motor innervation from spinal segments T3, T4, T5
- C. receives most of its sensory innervation from the intercostal nerves
- D. tends to flatten during inspiration and, thereby, constricts the vena cava
- E. all of the above (A-D)

The inguinal ligament:

E

- A. extends from the superior iliac crest to the pubic tubercle
- B. forms the floor of the inguinal canal
- C. represents the inferior border of the external abdominal oblique muscle
- D. after attaching to the pubic tubercle, curves inferiorly and posteriorly to form the lacunar and pectineal ligament
- E. all of the above (A-D)

814. The conjoined tendon (falx inguinalis) is formed by the:

D

- A. rectus abdominis muscle or its aponeurosis
- B. external abdominal oblique muscle or its aponeurosis
- C. both A and B
- D. neither A nor B

Inferior to the arcuate line, the posterior wall of the rectus sheath is composed of:

D

- A. aponeurosis of the external abdominal oblique muscle
- B. aponeurosis of the internal abdominal oblique muscle
- C. aponeurosis of the transversus abdominal oblique muscle
- D. transversalis fascia
- E. all of the above (A-D)

816. The following statements are true about indirect inguinal hernias except they:

C

- A. are associated with a patent processus vaginalis
- B. do not occur within the inguinal triangle
- C. break through the conjoined tendon to reach the superficial opening of the inguinal canal
- D. are insheathed by all three layers of spermatic fascia

817. The following statements are true about the inguinal ligament except it:

D

- A. forms the floor of the inguinal canal
- B. is part of the aponeurosis of the external abdominal oblique
- C. is the origin for some fibers of the internal abdominal oblique
- D. forms the suspensory ligament of the penis (clitoris) medially

818. The following statements are true about the cisterna chyli except it:

B

- A. receives lymph from nodes draining the posterior abdominal wall
- B. is in direct continuity with the inferior vena cava
- C. is a terminal dilation of the thoracic duct
- D. is generally located inferior to the aortic hiatus

795. Which rami are associated with the cervical region of the sympathetic trunk: B
- A. white ramus
 - B. gray ramus
 - C. both A and B
 - D. neither A nor B
- Position of the deep inguinal ring relative to the inferior epigastric vessels is: B
- A. medial
 - B. lateral
 - C. posterior
 - D. anterior
 - E. superior
797. Regarding the pancreas: C
- A. exocrine secretions are released into interstitial spaces
 - B. the pancreas is invested by a prominent and distinct connective tissue capsule
 - C. the pancreas is retroperitoneal except for the tail which is intraperitoneal
 - D. centroaciner cells appear acidophilic when stained by routine hematoxylin and eosin methods due to the high number of secretory vesicles
 - E. all of the above (A-D)
798. Pelvic splanchnics are found in the: C
- A. infradiaphragmatic foregut
 - B. midgut
 - C. hindgut
 - D. both A and C
 - E. all of the above (A-C)
799. Primary arterial supply of which region travels via the superior mesenteric artery: B
- A. infradiaphragmatic foregut
 - B. midgut
 - C. hindgut
 - D. both A and C
 - E. all of the above (A-C)
800. The following facts are true for the inguinal ligament: E
- A. it is an aponeurosis of the external abdominal oblique muscle
 - B. it forms the floor of the inguinal canal
 - C. it is continuous with the lacunar ligament
 - D. it extends from the anterior superior iliac spine to the pubic tubercle
 - E. all of the above (A-D)
801. Collateral circulation derived from extra-abdominal vessels supplies which region: D
- A. infradiaphragmatic foregut
 - B. midgut
 - C. hindgut
 - D. both A and C
 - E. all of the above (A-C)
- Which region is related to dorsal mesentery: E
- A. infradiaphragmatic foregut
 - B. midgut
 - C. hindgut
 - D. both A and C
 - E. all of the above (A-C)

778. The blood supply to the ureters is derived from which of the following: E
- A. renal artery
 - B. gonadal artery
 - C. inferior vesical artery
 - D. both A and B
 - E. all of the above (A-C)
779. Which of the following statements about the inguinal canal is incorrect: A
- A. the deep inguinal ring represents a partial interruption in the peritoneum
 - B. the anterior wall is formed by the aponeurosis of external abdominal oblique
 - C. the posterior wall is formed by transversalis fascia and reinforced medially by the conjoint tendon
 - D. an indirect inguinal hernia may result from a congenitally persistent processus vaginalis
- The vermiform appendix can be easily located by following the: C
- A. descending branch of the right colic artery
 - B. superior mesenteric vein
 - C. teniae coli of the ascending colon
 - D. descending branch of the left colic artery
781. Autonomic control of peristalsis in the descending colon should not be affected by bilateral lumbar sympathectomy because: A
- A. the descending colon is controlled chiefly by parasympathetic innervation from pelvic splanchnic nerves
 - B. the descending colon receives its parasympathetic innervation from thoracic splanchnic nerves
 - C. only presynaptic sympathetic fibers have been severed
782. An incision made parallel to the inguinal ligament to the depth of the aponeurosis of the external abdominal oblique muscle would include all of the following layers except: C
- A. deep fascia
 - B. Scarpa's fascia
 - C. transversalis fascia
 - D. Camper's fascia
- Pain associated with external hemorrhoids is mediated by: C
- A. the sacral sympathetic chain
 - B. pelvic splanchnic nerves
 - C. the pudendal nerves
 - D. the hypogastric nerve to the lumbar splanchnic nerves
- All of the following statements are true except: D
- A. the rectus sheath is formed by the aponeuroses of the internal abdominal oblique, external abdominal oblique and transverse abdominis muscles
 - B. the arcuate line represents the inferior free edge of the posterior leaf of the rectus sheath
 - C. the muscles of the anterior abdominal wall are innervated by the lower intercostal nerves
 - D. the superior epigastric artery arises from the bifurcation of the lowest intercostal artery
785. The renal fascia separates the paranephric and perinephric fat and limits the spread of infection from all of the following locations except: C
- A. lesser sac
 - B. suprarenal gland
 - C. deep pelvis
 - D. contralateral kidney

762. The numerous small lobules of fat enclosed in peritoneum around the large intestine are called: C
- A. haustra coli
 - B. tenia coli
 - C. epiploic appendages
 - D. tenia omentalis
 - E. colic pylori

Indicate the correct statement:

C

- A. the blood supply of the stomach is provided by branches of the portal vein in the greater curvature and spleen
- B. the lymph drainage of the stomach is directed toward pelvic lymph nodes
- C. the left and right gastric arteries, the left and right gastroepiploic arteries and the posterior gastric artery are the major arteries of the stomach
- D. the gastroduodenal artery is a short branch arising from the aorta just below the level of the renal arteries
- E. the gastroepiploic arteries anastomose in the lesser curvature, near the iliac artery

The lumbar splanchnic nerves:

C

- A. contain preganglionic sympathetic fibers
- B. originate from nerve cell bodies located in the intermediolateral cell column
- C. both A and B
- D. neither A nor B

The model or standard pylorus is the:

B

- A. pylorus ilealis
- B. gastroduodenal pylorus
- C. sigmoidorectal pylorus
- D. ileocolic pylorus
- E. biliopancreatic pylorus

766. Inflammation of the gallbladder may present with a symptom of pain in the skin of the triangles of the neck. The anatomical basis for this "referred pain" is: C
- A. parasympathetic fibers from the gallbladder and neck visceral all enter the central nervous system at upper thoracic levels
 - B. parasympathetic fibers from the gallbladder ascend to the cervical spinal cord to enter the central nervous system
 - C. sensory endings in the parietal peritoneum of the diaphragm are stimulated by contact with the diseased gallbladder and the afferent impulses are conducted in the phrenic nerve (C3-C5). The pain is sensed over the mid-cervical dermatomes
 - D. the wide distribution of the vagus nerve accounts for this referred pain. The vagus conducts pain impulses from all of the head, neck, thorax, and most of the abdomen
 - E. the anatomical basis for referred pain is unknown

Structures attached to the twelfth rib include all except the:

D

- A. diaphragm
- B. quadratus lumborum
- C. intercostal muscles
- D. psoas major
- E. lateral arcuate ligament

768. Nerves located along the medial side of psoas major muscle include all of the following except: E
- A. obturator
 - B. ventral ramus of L4
 - C. ventral ramus of L5
 - D. sympathetic trunk
 - E. nerve to piriformis

Which of the following does not contribute to boundaries of the posterior abdominal wall: E

- A. lumbar vertebral bodies
- B. twelfth rib
- C. psoas major muscle
- D. quadratus lumborum muscle
- E. internal abdominal oblique

747. In the midaxillary line the lowest point of the pleural cavity during quiet respiration lies deep to rib: C

- A. 8
- B. 9
- C. 10
- D. 11
- E. 12

The umbilical region of the anterior abdominal wall would include portions of which quadrants: E

- A. upper right and left quadrant
- B. lower right and left quadrant
- C. upper and lower right quadrant
- D. upper and lower left quadrant
- E. all of the above (A-D)

The majority of the definitive (adult) posterior parietal peritoneum of the infracolic portion of the peritoneal cavity is created by: D

- A. embryonic posterior parietal peritoneum
- B. the serous membrane (peritoneum) on the original right side of the dorsal mesentery
- C. the serous membrane (peritoneum) on the original left side of the dorsal mesentery
- D. both A and B
- E. all of the above (A-C)

A fusion fascia is formed: D

- A. posterior to most of the pancreas
- B. between the inferior mesenteric vein and the left testicular vein
- C. in the bare area of the liver
- D. both A and B
- E. all of the above (A-C)

751. The vertebrocostal triangle: C

- A. is a frequent site of hiatal herniation
- B. is located on the left side of the diaphragm at the level of the twelfth rib
- C. both A and B
- D. neither A nor B

752. The superior hypogastric plexus transmits which type of nerve fibers: D

- A. parasympathetic
- B. sympathetic
- C. visceral afferents
- D. both B and C
- E. all of the above (A-C)

All of the following statements are true except: C

- A. access to the lesser sac is through the epiploic foramen
- B. the left colicophrenic ligament closes the left colic gutter
- C. the lateral umbilical (inguinal) fossa is the site of an indirect hernia
- D. mesenteries are formed by back to back layers of peritoneum

730. The lesser omentum reaches the lesser curvature of the stomach and is attached to the: C
- A. caudate and quadrate lobe
 - B. vena cava notch and ligamentum teres
 - C. porta hepatis and fossa of round ligament
 - D. gallbladder notch and venous ligament
 - E. porta hepatis and fossa of ligamentum venosum
731. The transversus abdominis muscle has its insertions: C
- A. on the linea alba and the pubic crest
 - B. on the linea alba only
 - C. on the linea alba, pubic crest and pecten pubis
 - D. only on the pecten pubis and pubic crest
 - E. on the pubic tubercle
732. The rectus abdominis muscle is innervated by the: B
- A. inferior five intercostal nerves, subcostal nerve and ilioinguinal nerve
 - B. inferior five intercostal nerves and subcostal nerve
 - C. inferior five intercostal nerves, subcostal nerve, iliohypogastric and ilioinguinal nerves
 - D. subcostal nerve only
 - E. iliohypogastric and ilioinguinal nerves
- The external oblique muscle originates from the: B
- A. internal surface of the inferior six costal cartilages
 - B. external surface of ribs 5-12
 - C. anterior surface of the xiphoid process and the 5th to 7th costal cartilages
 - D. inferior borders of the inferior three or four ribs
 - E. external surface of the lower 4 ribs
- The posterior wall of the rectus sheath superior to the arcuate line is formed by the: C
- A. aponeurosis of the internal oblique and external oblique muscles
 - B. aponeurosis of the external oblique muscle
 - C. aponeurosis of the transversus abdominis and internus oblique muscle
 - D. aponeurosis of the transversus abdominis and external oblique muscle
 - E. transversalis fascia
735. The psoas major muscle: B
- A. has its origin from the lower three thoracic and all lumbar vertebral bodies
 - B. contains the lumbar plexus
 - C. is covered on its anterior surface by fascia which is continuous with Scarpa's fascia of the anterior abdominal wall
 - D. inserts on the pecten pubis
 - E. is associated with the median arcuate ligament
- Select the correct statement: B
- A. falciform ligament actively suspends the liver
 - B. the liver occupies the right hypochondriac and epigastric region of abdomen
 - C. the quadrate lobe of liver is adjacent to the inferior vena cava
 - D. the caudate lobe is the bed for the gallbladder
 - E. the cystic bile duct leads into the proximal duodenum
- The superficial inguinal ring is the aperture in the: D
- A. aponeurosis of the external oblique and internal oblique muscles
 - B. conjoint tendon
 - C. transversalis fascia
 - D. aponeurosis of the external oblique muscle
 - E. linea alba

714. The following are true concerning the diaphragm except: B
- A. takes some origin from the vertebral column in the form of crura
 - B. facilitates inspiration by decreasing the diameter of the thorax upon contraction
 - C. has an opening at the level of T₁₀ through which the esophagus passes
 - D. is a common site for congenital hernias when it fails to develop properly
- The following are true about vessels of the posterior abdominal wall except: A
- A. the inferior vena cava is formed by the union of the splenic and superior mesenteric veins
 - B. the aorta enters the abdomen at the level of T₁₂ under the median arcuate ligament
 - C. the median sacral artery arises at the bifurcation of the aorta and descends in front of the sacrum and coccyx
 - D. the paired lumbar arteries contribute blood supply to the posterior abdominal wall
716. Select the one best answer: C
- A. the superior epigastric artery is a branch of the external iliac artery
 - B. the anterior portion of the rectus sheath is formed by the aponeuroses of the external abdominal oblique and the transversus abdominis muscle
 - C. the inguinal canal terminates at the superficial inguinal ring (a defect in the external abdominal oblique muscle)
 - D. the conjoint tendon consists of the lower fibers of the external abdominal oblique muscle and the inguinal ligament
717. Which abdominal structure gives rise to (is continuous with) the external spermatic fascia following descent of the testes during development: A
- A. external abdominal oblique aponeurosis
 - B. transversalis fascia
 - C. transversus abdominis muscle
 - D. peritoneum
 - E. internal abdominal oblique
718. Which of the following is/are true regarding the nerves of the posterior abdominal wall: B
- A. the thoracic splanchnic nerves contribute parasympathetic components to the prevertebral plexus
 - B. the superior hypogastric plexus is a continuation of the aortic plexus below the bifurcation of the aorta
 - C. both A and B
 - D. neither A nor B
- Which abdominal structure gives rise to (is continuous with) the cremasteric fascia (muscle) following descent of the testes during development: E
- A. external abdominal oblique aponeurosis
 - B. transversalis fascia
 - C. transversus abdominis muscle
 - D. peritoneum
 - E. internal abdominal oblique
- Dilator muscles of the alimentary canal are made up of: A
- A. longitudinal muscle fibers
 - B. elliptical muscle fibers
 - C. circular muscle fibers
 - D. plexiform muscle fibers
 - E. orbicular fibers
721. The diaphragm: D
- A. receives sensory innervation via the thoracic splanchnic nerves
 - B. is composed only of muscle fibers
 - C. both A and B
 - D. neither A nor B

697. The gallbladder receives blood from the: D
A. superior mesenteric artery
B. middle hepatic artery and duodenal artery
C. left hepatic artery
D. cystic artery
E. cystic artery and right renal artery
698. The left kidney: A
A. is posterior to the tail of the pancreas
B. is completely covered by parietal peritoneum on its anterior surface
C. both A and B
D. neither A nor B
699. The human kidney is considered: C
A. unilobar and multilobular
B. multilobar and unilobular
C. multilobar and multilobular
D. unilobar and unilobular
700. Which of the following is not true in relation to the stomach: C
A. it is located in the supramesocolic compartment
B. the right gastroepiploic artery, running along the greater curvature, is a branch of the gastroduodenal artery
C. the cardiac notch is an incisure found at the most dependent point of the lesser curvature
D. the greater curvature gives attachment to the greater omentum
701. The grey rami communicants: A
A. are postganglionic sympathetic fibers
B. are preganglionic sympathetic fibers
C. both A and B
D. neither A nor B
702. The inguinal ligament is: D
A. formed by the lower border of the aponeurosis of the external oblique muscle
B. attached laterally and superiorly to the anterior superior iliac spine
C. attached inferiorly and medially to the iliac tubercle
D. both A and B
E. all of the above (A-C)
703. Surgical mobilization of the descending colon from the posterior abdominal wall can be accomplished by reestablishing a mesentery for this section of the bowel. This reestablished portion of mesentery would contain: E
A. left lumbar arteries
B. left renal artery
C. middle colic artery
D. left gonadal artery
E. none of the above (A-D)
704. The common bile duct is divided into: E
A. omental, retroduodenal and prepancreatic portions
B. omental and retroduodenal portions
C. omental, retroduodenal and intraduodenal portions
D. omental, intraparietal and retroduodenal portions
E. omental, retroduodenal, retropancreatic, and intraduodenal portions